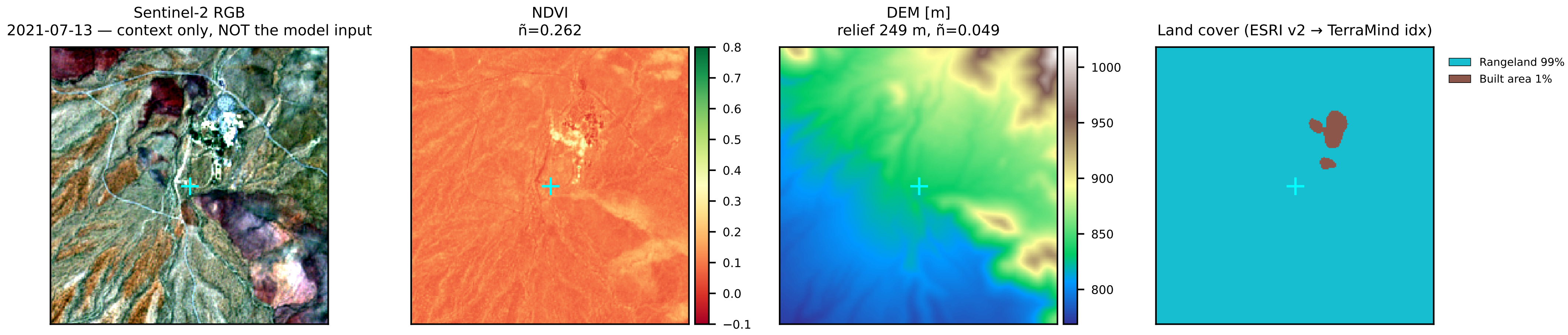
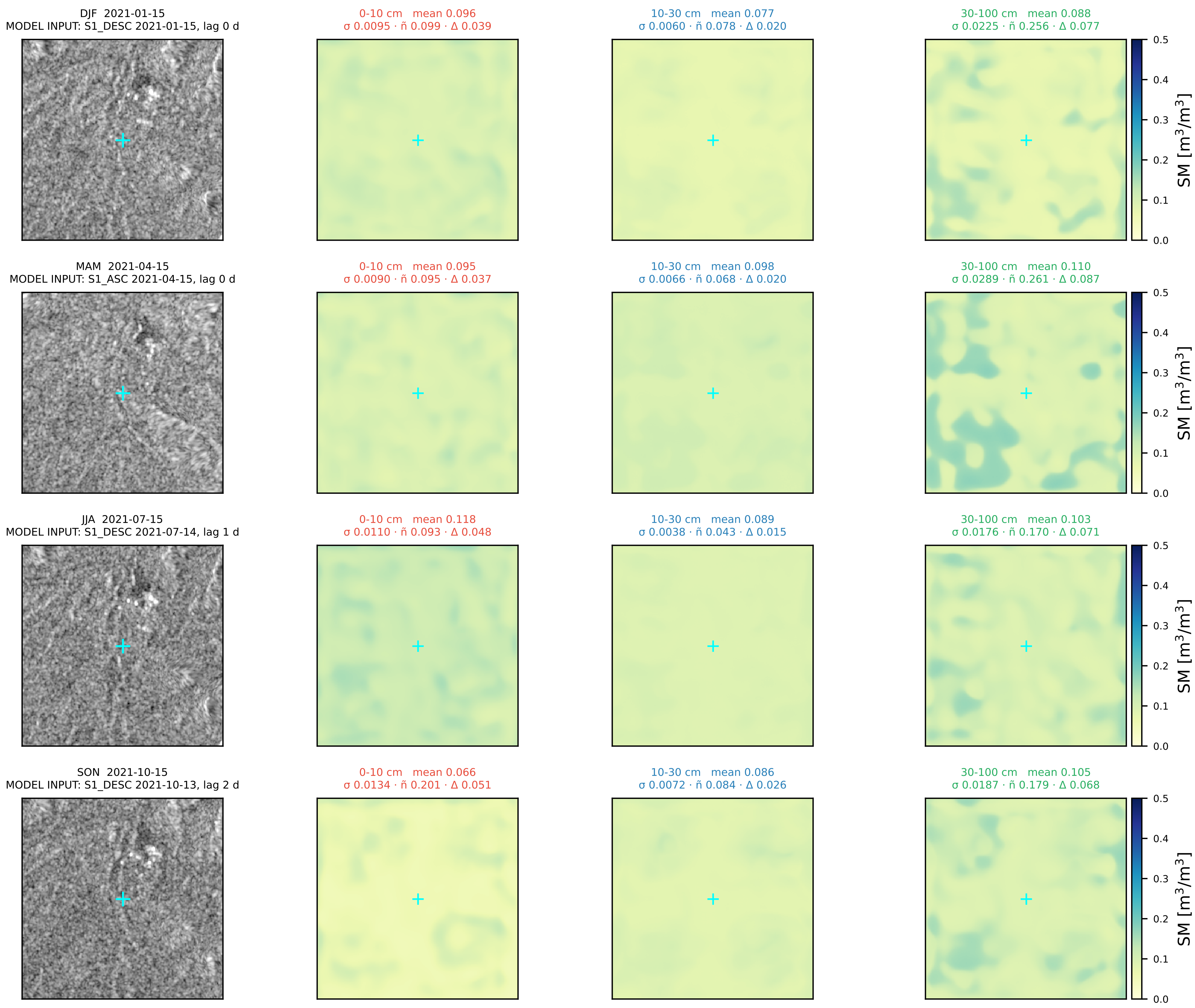
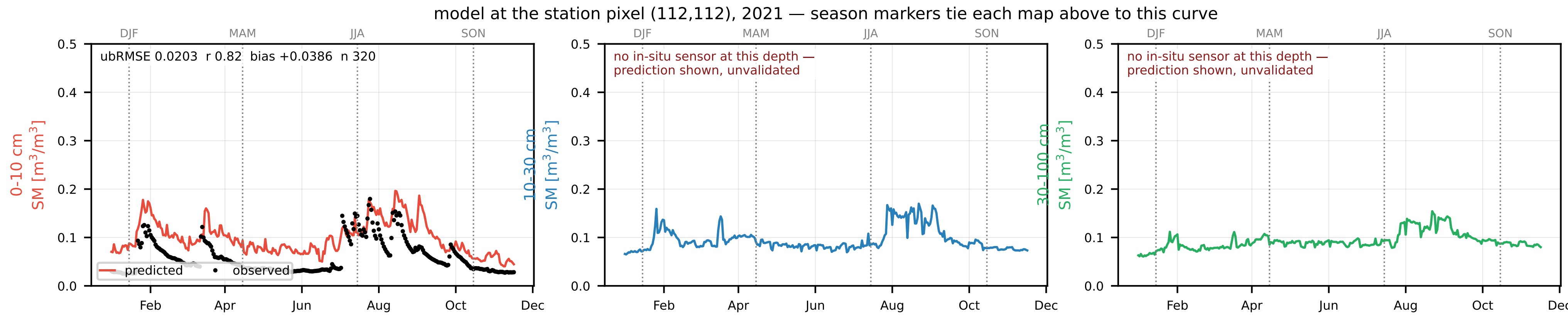
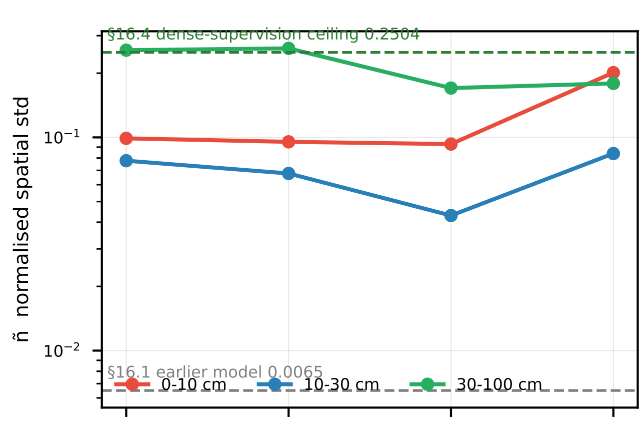
§23 Spatial and temporal validation — ISMN_USCRN_Tucson-11-W [best_ubrmse]
USCRN network · 32.2395°N, -111.1696°E · 833 m a.s.l. · IGBP SAV (savanna) · Köppen BWh (hot desert) · 8 yr record
0-10 cm ubRMSE 0.0219





model at the station pixel (112,112), 2021 — season markers tie each map above to this curve



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SYMBOLS
σ        spatial standard deviation of the predicted map [m³/m³]
σ_time   temporal std of the station-pixel prediction, this year
ñ        normalised spatial std = σ / |mean of map|  [−]
Δ        spread of the map, 98th minus 2nd percentile [m³/m³]
r        Pearson correlation coefficient  [−]
14²      14×14 token grid = model bottleneck, ≈160 m pixels
var kept at 14²   share of map variance surviving
                  block-averaging to that grid
r(season~season)  similarity of the spatial pattern between
                  seasons: 1 = one fixed pattern, 0 = unrelated
anchor, lag       the satellite acquisition the model actually
                  consumed, and its age in days
ubRMSE   unbiased RMSE = √(MSE − bias²)  [m³/m³]
SM       volumetric soil moisture  [m³/m³]
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SPATIAL SUMMARY

  0-10 cm  σ 0.0107  σ_time 0.0338  σ/σ_time 0.32
           r(DEM,14²) -0.04   r(NDVI,14²) -0.00
           var kept at 14² 64%   r(season~season) +0.38
 10-30 cm  σ 0.0059  σ_time nan  σ/σ_time nan
           r(DEM,14²) -0.13   r(NDVI,14²) -0.09
           var kept at 14² 77%   r(season~season) +0.65
30-100 cm  σ 0.0219  σ_time nan  σ/σ_time nan
           r(DEM,14²) -0.21   r(NDVI,14²) -0.13
           var kept at 14² 70%   r(season~season) +0.58

cross-depth r(0-10, 30-100) -0.01 — architectural
(star residual, model.py:276), not learned depth structure.

HOW TO READ
r(DEM/NDVI) ≈ 0  → the structure is not the terrain or
                   the vegetation of this footprint
r(season~season) → 1  → one pattern, mean-shifted
var kept at 14² ≈ 100%  → no detail below ≈160 m
σ/σ_time ≈ 1  → across 2.2 km the model varies as much
                as it does across the whole year
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Predicted soil moisture for ISMN_USCRN_Tucson-11-W (OOS split, checkpoint cls_depth_star_reg epoch 16), year 2021. Top: what the 2.24 × 2.24 km footprint contains — Sentinel-2 RGB, NDVI, terrain and ESRI land cover (stored as TerraMind indices; ESRI v1 'Grass' and 'Scrub/Shrub' are merged into Rangeland). Middle: the acquisition the model actually consumed and the predicted 224 × 224 field at each depth, all on one 0–0.5 m³/m³ scale; the anchor may pre-date the prediction by up to 364 days, so its lag is stated. Then: the model's own time series at the station pixel for the same year, with the four map dates marked. Bottom: how much the field varies in space, against the two reference levels of §16.1 and §16.4, the symbol list, and whether the spatial variation corresponds to anything physical. The cyan cross marks the station pixel (112,112) — the only pixel the training loss ever sees.